

LMx37 3-Terminal Adjustable Regulators

1 Features

- Output Voltage Range Adjustable From -1.2 V to -37 V
- Output Current Capability of 1.5 A Max
- Input Regulation Typically 0.01% Per Input-Voltage Change
- Output Regulation Typically 0.3%
- Peak Output Current Constant Over Temperature Range of Regulator
- Ripple Rejection Typically 77 dB
- Direct Replacement for Industry-Standard LM237 and LM337

2 Applications

- Applications Requiring Negative Output Voltage or Precision Current Regulation
- Consumer Electronics
- End Equipment
- Portable Applications

4 Simplified Schematic

3 Description

The LM237 and LM337 are adjustable 3-terminal negative-voltage regulators capable of supplying in excess of -1.5 A over an output voltage range of -1.2 V to -37 V . They require only two external resistors to set the output voltage and one output capacitor for frequency compensation.

Device Information⁽¹⁾

PART NUMBER	PACKAGE	BODY SIZE (NOM)
LMx37	TO-220 (4)	10.16 mm x 8.82 mm
	TO-263 (4)	10.16 mm x 9.02 mm
	TO-252 (4)	6.6 mm x 6.10 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

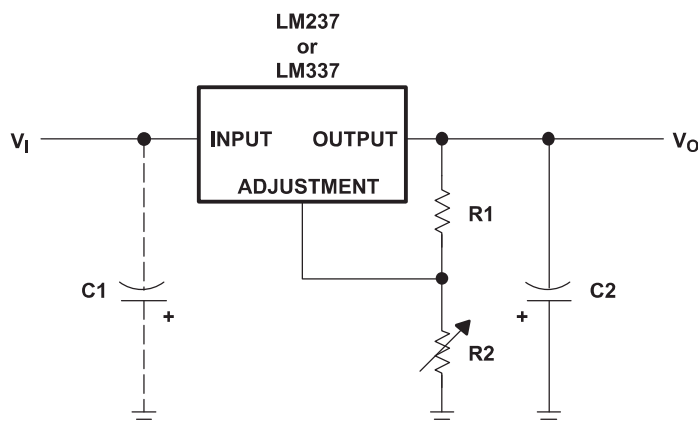


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5 Revision History

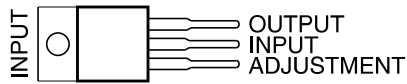
Changes from Revision K (November 2007) to Revision L

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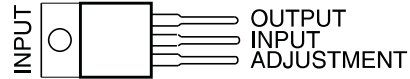
• Added <i>Applications</i> , <i>Device Information</i> table, <i>Pin Functions</i> table, <i>ESD Ratings</i> table, <i>Thermal Information</i> table, <i>Typical Characteristics</i> , <i>Feature Description</i> section, <i>Device Functional Modes</i> , <i>Application and Implementation</i> section, <i>Power Supply Recommendations</i> section, <i>Layout</i> section, <i>Device and Documentation Support</i> section, and <i>Mechanical, Packaging, and Orderable Information</i> section.	1
• Deleted <i>Ordering Information</i> table.	1

6 Pin Configuration and Functions

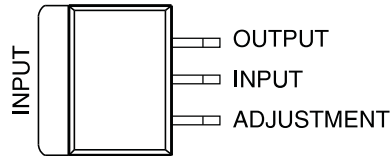
**LM237, LM337...KC (TO-220) PACKAGE
(TOP VIEW)**



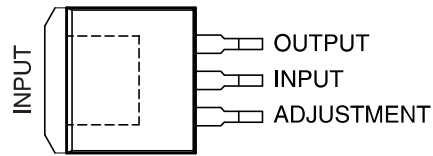
**LM337...KCS (TO-220) PACKAGE
(TOP VIEW)**



**LM337...KTE, KTP, OR KVV PACKAGE
(TOP VIEW)**



**LM337...KTT (TO-263) PACKAGE
(TOP VIEW)**



Pin Functions

PIN		TYPE	DESCRIPTION
NAME	NO.		
ADJUSTMENT	1	I	Adjustment pin for the output voltage. Connect two external resistors to adjust the output voltage.
INPUT	2	I	Input voltage. The input voltage and current will be designated V_I and I_I respectively.
OUTPUT	3	O	Output voltage. The output voltage and current will be designated V_O and I_O respectively.

LM237, LM337

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7 Specifications

7.1 Absolute Maximum Ratings

over operating temperature ranges (unless otherwise noted)⁽¹⁾

		MIN	MAX	UNIT
$V_I - V_O$	Input-to-output differential voltage		-40	V
T_J	Operating virtual junction temperature		150	°C
	Lead temperature	1.6 mm (1/16 in) from case for 10 s	260	°C
T_{stg}	Storage temperature range	-65	150	°C

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

7.2 ESD Ratings

		VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human body model (HBM), per ANSI/ESDA/JEDEC JS-001, all pins ⁽¹⁾	1500
		Charged device model (CDM), per JEDEC specification JESD22-C101, all pins ⁽²⁾	1500
			V

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

7.3 Recommended Operating Conditions

		MIN	MAX	UNIT
$V_I - V_O$	Input-to-output differential voltage	-2.5	-37	
I_O	Output current	$ V_I - V_O \leq 40 \text{ V}, P \leq 15 \text{ W}$	10	1500
		$ V_I - V_O \leq 10 \text{ V}, P \leq 15 \text{ W}$	6	1500
T_J	Operating virtual junction temperature	LM237	-25	150
		LM337	0	125

7.4 Thermal Information

THERMAL METRIC ⁽¹⁾		LM237	LMx37		LM337			UNIT
		KC	KCS	KTE	KTP	KTT	KVU	
		4 PINS	4 PINS	4 PINS	4 PINS	4 PINS	4 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	24.8	24.8	23	28	25.3	30.3	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	3	3	3	19	30.3	N/A	

- (1) For more information about traditional and new thermal metrics, see the *IC Package Thermal Metrics* application report (SPRA953).

7.5 Electrical Characteristics

over recommended ranges of operating virtual junction temperature (unless otherwise noted)

PARAMETER	TEST CONDITIONS ⁽¹⁾		LM237			LM337			UNIT
			MIN	TYP	MAX	MIN	TYP	MAX	
Input regulation ⁽²⁾	$V_I - V_O = -3 \text{ V to } -40 \text{ V}$	$T_J = 25^\circ\text{C}$		0.01	0.02		0.01	0.04	%V
		$T_J = \text{MIN to MAX}$		0.02	0.05		0.02	0.07	
Ripple rejection	$V_O = -10 \text{ V, } f = 120 \text{ Hz}$			60			60		dB
	$V_O = -10 \text{ V, } f = 120 \text{ Hz, } C_{\text{ADJ}} = 10 \mu\text{F}$		66	77		66	77		
Output regulation	$I_O = 10 \text{ mA to } 1.5 \text{ A, } T_J = 25^\circ\text{C}$	$ V_O \leq 5 \text{ V}$			25			50	mV
		$ V_O \geq 5 \text{ V}$		0.3%	0.5%		0.3%	1%	—
	$I_O = 10 \text{ mA to } 1.5 \text{ A}$	$ V_O \leq 5 \text{ V}$			50			70	mV
		$ V_O \geq 5 \text{ V}$			1%			1.5%	—
Output-voltage change with temperature	$T_J = \text{MIN to MAX}$			0.6%			0.6%		—
Output-voltage long-term drift	After 1000 h at $T_J = \text{MAX}$ and $V_I - V_O = -40 \text{ V}$			0.3%	1%		0.3%	1%	—
Output noise voltage	$f = 10 \text{ Hz to } 10 \text{ kHz, } T_J = 25^\circ\text{C}$			0.003%			0.003%		—
Minimum output current to maintain regulation	$ V_I - V_O \leq 40 \text{ V}$			2.5	5		2.5	10	mA
	$ V_I - V_O \leq 10 \text{ V}$			1.2	3		1.5	6	
Peak output current	$ V_I - V_O \leq 15 \text{ V}$		1.5	2.2		1.5	2.2		A
	$ V_I - V_O \leq 40 \text{ V, } T_J = 25^\circ\text{C}$		0.24	0.4		0.15	0.4		
ADJUSTMENT current				65	100		65	100	μA
Change in ADJUSTMENT current	$V_I - V_O = -2.5 \text{ V to } -40 \text{ V, } I_O = 10 \text{ mA to MAX, } T_J = 25^\circ\text{C}$			2	5		2	5	μA
Reference voltage (OUTPUT to ADJUSTMENT)	$V_I - V_O = -3 \text{ V to } -40 \text{ V, } I_O = 10 \text{ mA to } 1.5 \text{ A, } P \leq \text{rated dissipation}$	$T_J = 25^\circ\text{C}$	-1.225	-1.25	-1.275	-1.213	-1.25	-1.287	V
		$T_J = \text{MIN to MAX}$	-1.2	-1.25	-1.3	-1.2	-1.25	-1.3	
Thermal regulation	Initial $T_J = 25^\circ\text{C, } 10\text{-ms pulse}$			0.002	0.02		0.003	0.04	%/W

- (1) Unless otherwise noted, the following test conditions apply: $|V_I - V_O| = 5 \text{ V}$ and $I_O = 0.5 \text{ A}$. For conditions shown as MIN or MAX, use the appropriate value specified under recommended operating conditions. All characteristics are measured with a $0.1\text{-}\mu\text{F}$ capacitor across the input and a $1\text{-}\mu\text{F}$ capacitor across the output. Pulse-testing techniques are used to maintain the junction temperature as close to the ambient temperature as possible. Thermal effects must be taken into account separately.
- (2) Input regulation is expressed here as the percentage change in output voltage per 1-V change at the input.

7.6 Electrical Characteristics

$T_J = 25^\circ\text{C}$

PARAMETER	TEST CONDITIONS ⁽¹⁾		LM237, LM337			UNIT
			MIN	TYP	MAX	
Input regulation ⁽²⁾	$V_I - V_O = -3 \text{ V to } -40 \text{ V}$			0.01	0.04	%/V
Ripple rejection	$V_O = -10 \text{ V, } f = 120 \text{ Hz}$			60		dB
	$V_O = -10 \text{ V, } f = 120 \text{ Hz, } C_{\text{ADJ}} = 10 \mu\text{F}$		66	77		
Output regulation	$I_O = 10 \text{ mA to } 1.5 \text{ A}$	$ V_O \leq 5 \text{ V}$			50	mV
		$ V_O \geq 5 \text{ V}$		0.3%	1%	—
Output noise voltage	$f = 10 \text{ Hz to } 10 \text{ kHz}$			0.003%		—
Minimum output current to maintain regulation	$ V_I - V_O \leq 40 \text{ V}$			2.5	10	mA
	$ V_I - V_O \leq 10 \text{ V}$			1.5	6	
Peak output current	$ V_I - V_O \leq 15 \text{ V}$		1.5	2.2		A
	$ V_I - V_O \leq 40 \text{ V}$		0.15	0.4		
ADJUSTMENT current				65	100	μA
Change in ADJUSTMENT current	$V_I - V_O = -2.5 \text{ V to } -40 \text{ V, } I_O = 10 \text{ mA to MAX}$			2	5	μA
Reference voltage (OUTPUT to ADJUSTMENT)	$V_I - V_O = -3 \text{ V to } -40 \text{ V, } I_O = 10 \text{ mA to } 1.5 \text{ A, } P \leq \text{rated dissipation}$		-1.213	-1.25	-1.287	V

- (1) Unless otherwise noted, the following test conditions apply: $|V_I - V_O| = 5 \text{ V}$ and $I_O = 0.5 \text{ A}$. All characteristics are measured with a $0.1\text{-}\mu\text{F}$ capacitor across the input and a $1\text{-}\mu\text{F}$ capacitor across the output. Pulse-testing techniques are used to maintain the junction temperature as close to the ambient temperature as possible. Thermal effects must be taken into account separately.
- (2) Input regulation is expressed here as the percentage change in output voltage per 1-V change at the input.

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7.7 Typical Characteristics

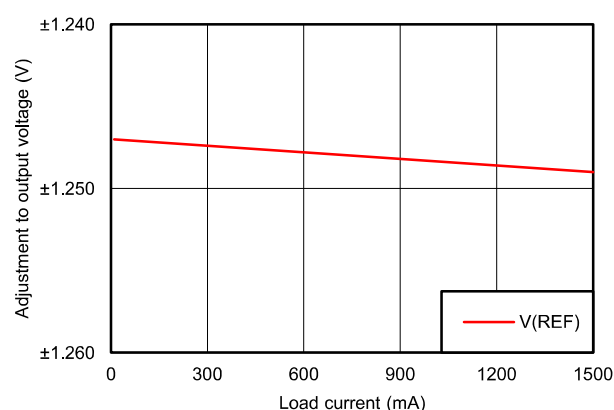


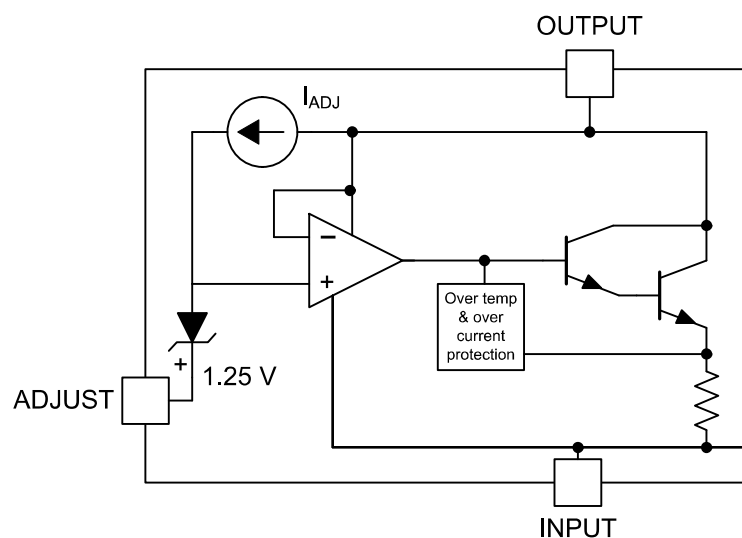
Figure 1. Adjustment Voltage vs Load current ($V_{IN} = -4.3$ V)

8 Detailed Description

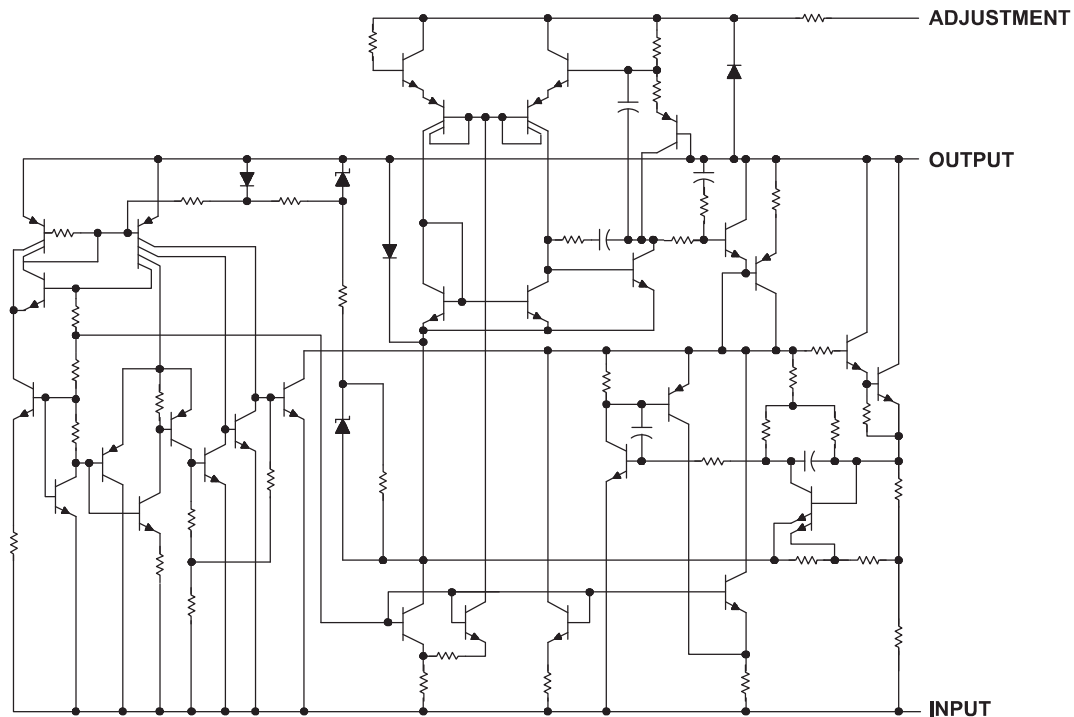
8.1 Overview

The LMx37 devices are adjustable 3-terminal negative-voltage regulators capable of supplying in excess of -1.5 A over an output voltage range of -1.2 V to -37 V. They are exceptionally easy to use, requiring only two external resistors to set the output voltage and one output capacitor for frequency compensation. The current design is optimized for excellent regulation and low thermal transients. In addition, LM237 and LM337 feature internal current limiting, thermal shutdown, and safe-area compensation, making them virtually immune to failure by overloads. The LMx37 devices serve a wide variety of applications, including local on-card regulation, programmable output-voltage regulation, and precision current regulation.

8.2 Functional Block Diagram



8.3 Design Schematic



8.4 Feature Description

8.4.1 Output Voltage Adjustment

The ADJUSTMENT pin serves as a voltage adjustment reference for the output. The ADJUSTMENT pin can be attached to a resistor divider circuit to adjust its own voltage level. The reference voltage $V_{\text{ADJUSTMENT}}$ will typically be 1.25 V higher than V_O .

8.5 Device Functional Modes

8.5.1 Adjustable Output Mode

The device has a single functional mode: Adjustable output voltage mode. A resistor divider circuit on the ADJUSTMENT pin determines the output voltage.

9 Application and Implementation

NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

9.1 Application Information

9.1.1 General Configurations

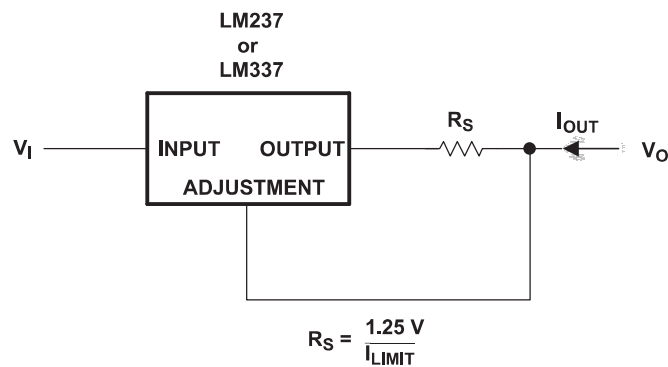


Figure 2. Current-Limiting Circuit

This application uses the LMx37 device's reference voltage, combined with the series resistor R_S , to limit the current to $1.25 \text{ V} \div R_S$.

9.2 Typical Application

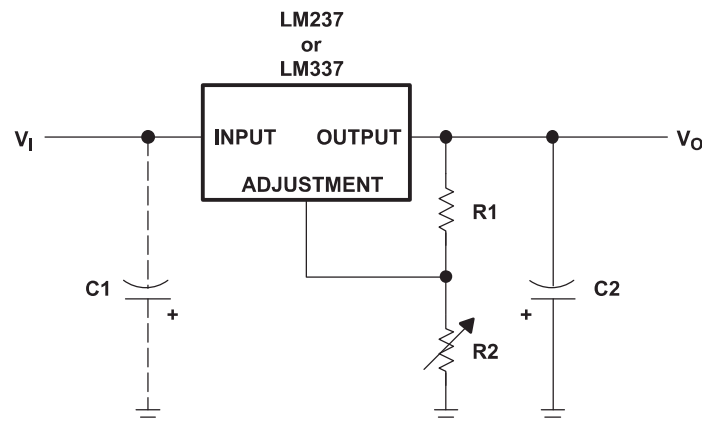


Figure 3. Adjustable Negative-Voltage Regulator

9.2.1 Design Requirements

- 1- μF solid tantalum on the input pin if the regulator is more than 10 cm from the power supply filter capacitor
- 1- μF solid tantalum or 10- μF aluminum electrolytic capacitor is required on the output pin for stability.
- R_1 , which is usually 120 Ω as part of the resistor divider.
- R_2 , which can be varied to change the value of V_O .

Typical Application (continued)

9.2.2 Detailed Design Procedure

V_O is determined by the values of R_1 and R_2 . Choosing $R_1 = 120\ \Omega$ means that about 10.42 mA of current will flow through R_1 . The ~10 mA of current satisfies the minimum operating current and renders I_{REF} negligible. Since the current is coming from ground, the same amount of current will flow through R_2 . Therefore, the size of R_2 will be the dominant factor in adjusting V_O . The relationship between R_1 , R_2 , and V_O is as follows:

$$R_2 = R_1 \left(\frac{V_O}{-1.25} - 1 \right)$$

where V_O is the output in volts.

(1)

9.2.3 Application Curves

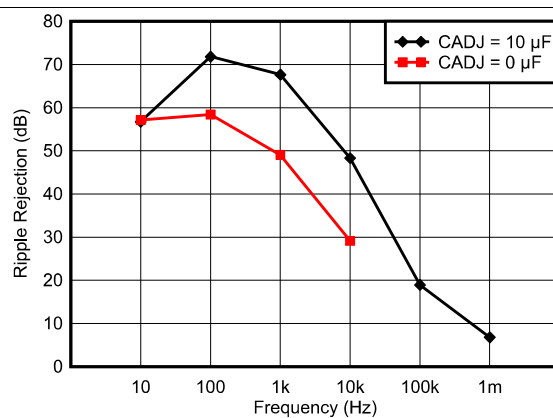


Figure 4. Ripple Rejection Over Frequency

D001

10 Power Supply Recommendations

For best performance, the difference in voltage between the output and input must be between -2.5 V and -37 V. A 1-μF solid tantalum capacitor is required on the input pin if the regulator is more than 10 cm from the power supply filter capacitor. A 1-μF solid tantalum or 10-μF aluminum electrolytic capacitor is required on the output pin for stability.

11 Layout

11.1 Layout Guidelines

Traces on the input and output pins should be thick enough to carry 1.5 A of current without violating thermal requirements of the device or the system. In addition, a 1- μ F solid tantalum capacitor is required on the input pin if the regulator is more than 10 cm from the power supply filter capacitor. A 1- μ F solid tantalum or 10- μ F aluminum electrolytic capacitor is required on the output pin for stability.

11.2 Layout Example

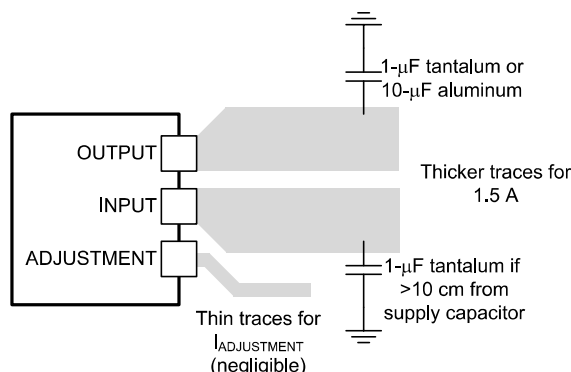


Figure 5. Layout Diagram

12 Device and Documentation Support

12.1 Related Links

The table below lists quick access links. Categories include technical documents, support and community resources, tools and software, and quick access to sample or buy.

Table 1. Related Links

PARTS	PRODUCT FOLDER	SAMPLE & BUY	TECHNICAL DOCUMENTS	TOOLS & SOFTWARE	SUPPORT & COMMUNITY
LM237	Click here	Click here	Click here	Click here	Click here
LM337	Click here	Click here	Click here	Click here	Click here

12.2 Trademarks

12.3 Electrostatic Discharge Caution



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

12.4 Glossary

[SLYZ022](#) — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
LM237KCSE3	Active	Production	TO-220 (KCS) 3	50 TUBE	Yes	SN	N/A for Pkg Type	-25 to 150	LM237
LM237KCSE3.A	Active	Production	TO-220 (KCS) 3	50 TUBE	Yes	SN	N/A for Pkg Type	-25 to 150	LM237
LM337KCSE3	Active	Production	TO-220 (KCS) 3	50 TUBE	Yes	SN	N/A for Pkg Type	0 to 125	LM337
LM337KCSE3.A	Active	Production	TO-220 (KCS) 3	50 TUBE	Yes	SN	N/A for Pkg Type	0 to 125	LM337
LM337KTTR	Active	Production	DDPAK/ TO-263 (KTT) 3	500 LARGE T&R	Yes	SN	Level-3-245C-168 HR	0 to 125	LM337
LM337KTTR.A	Active	Production	DDPAK/ TO-263 (KTT) 3	500 LARGE T&R	Yes	SN	Level-3-245C-168 HR	0 to 125	LM337
LM337KTTRG3	Active	Production	DDPAK/ TO-263 (KTT) 3	500 LARGE T&R	Yes	SN	Level-3-245C-168 HR	0 to 125	LM337
LM337KVURG3	Active	Production	TO-252 (KVU) 3	2500 LARGE T&R	Yes	SN	Level-3-260C-168 HR	0 to 125	LM337
LM337KVURG3.A	Active	Production	TO-252 (KVU) 3	2500 LARGE T&R	Yes	SN	Level-3-260C-168 HR	0 to 125	LM337

(1) **Status:** For more details on status, see our [product life cycle](#).

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

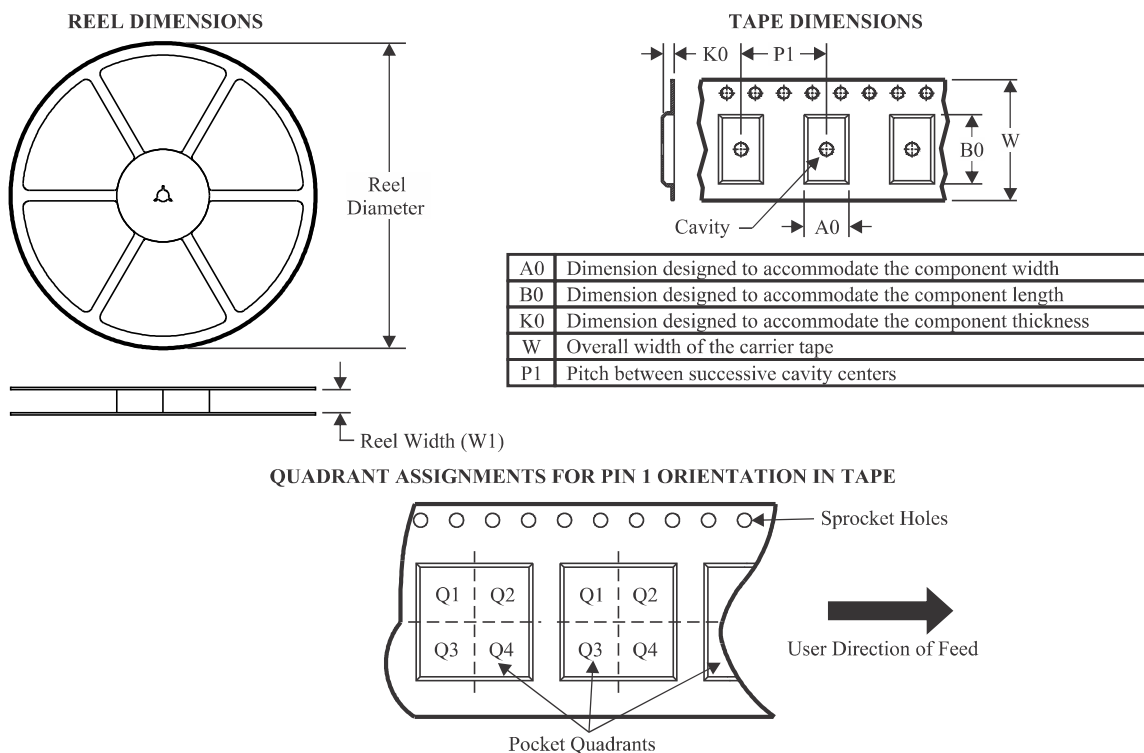
(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "-" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

Important Information and Disclaimer: The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

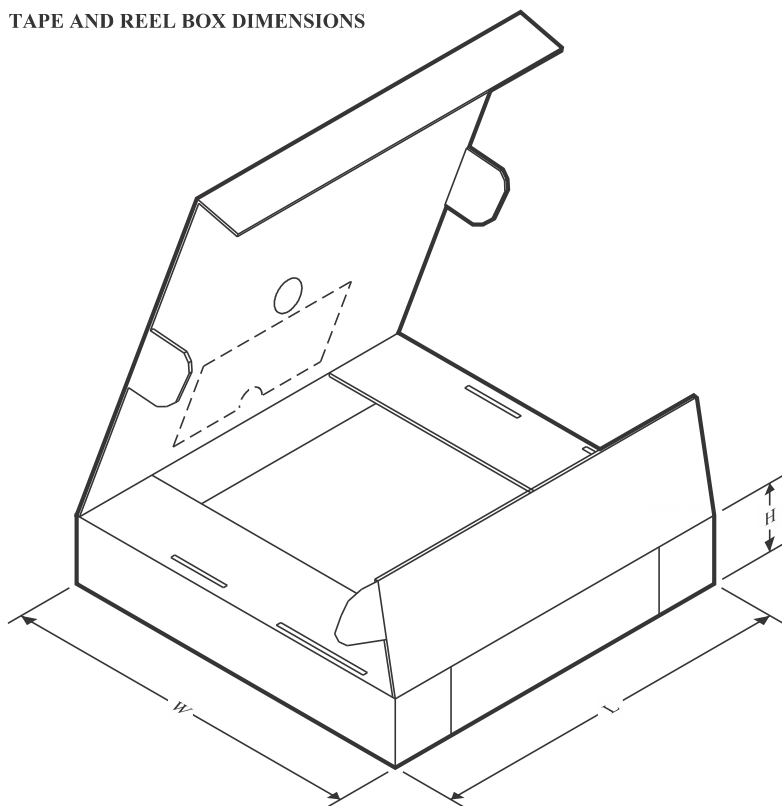
In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

TAPE AND REEL INFORMATION


*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
LM337KTTR	DDPAK/ TO-263	KTT	3	500	330.0	24.4	10.8	16.3	5.11	16.0	24.0	Q2
LM337KVURG3	TO-252	KVU	3	2500	330.0	16.4	6.9	10.5	2.7	8.0	16.0	Q2

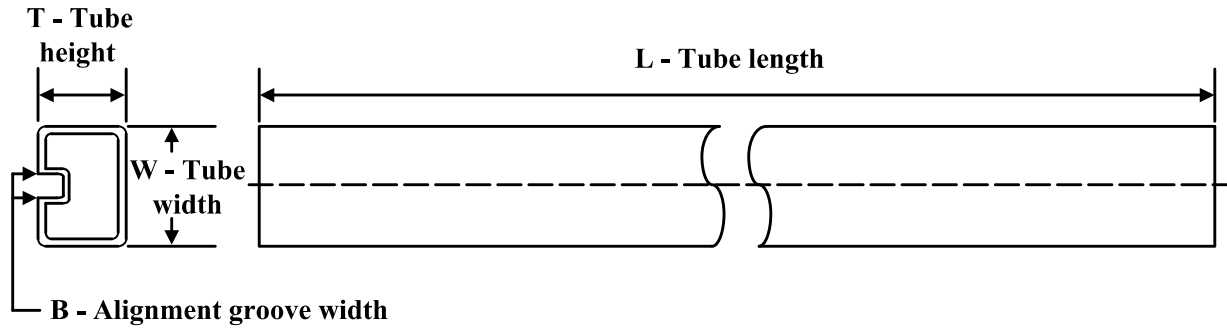
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
LM337KTTR	DDPAK/TO-263	KTT	3	500	340.0	340.0	38.0
LM337KVURG3	TO-252	KVU	3	2500	340.0	340.0	38.0

TUBE



*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
LM237KCSE3	KCS	TO-220	3	50	532	34.1	700	9.6
LM237KCSE3.A	KCS	TO-220	3	50	532	34.1	700	9.6
LM337KCSE3	KCS	TO-220	3	50	532	34.1	700	9.6
LM337KCSE3.A	KCS	TO-220	3	50	532	34.1	700	9.6

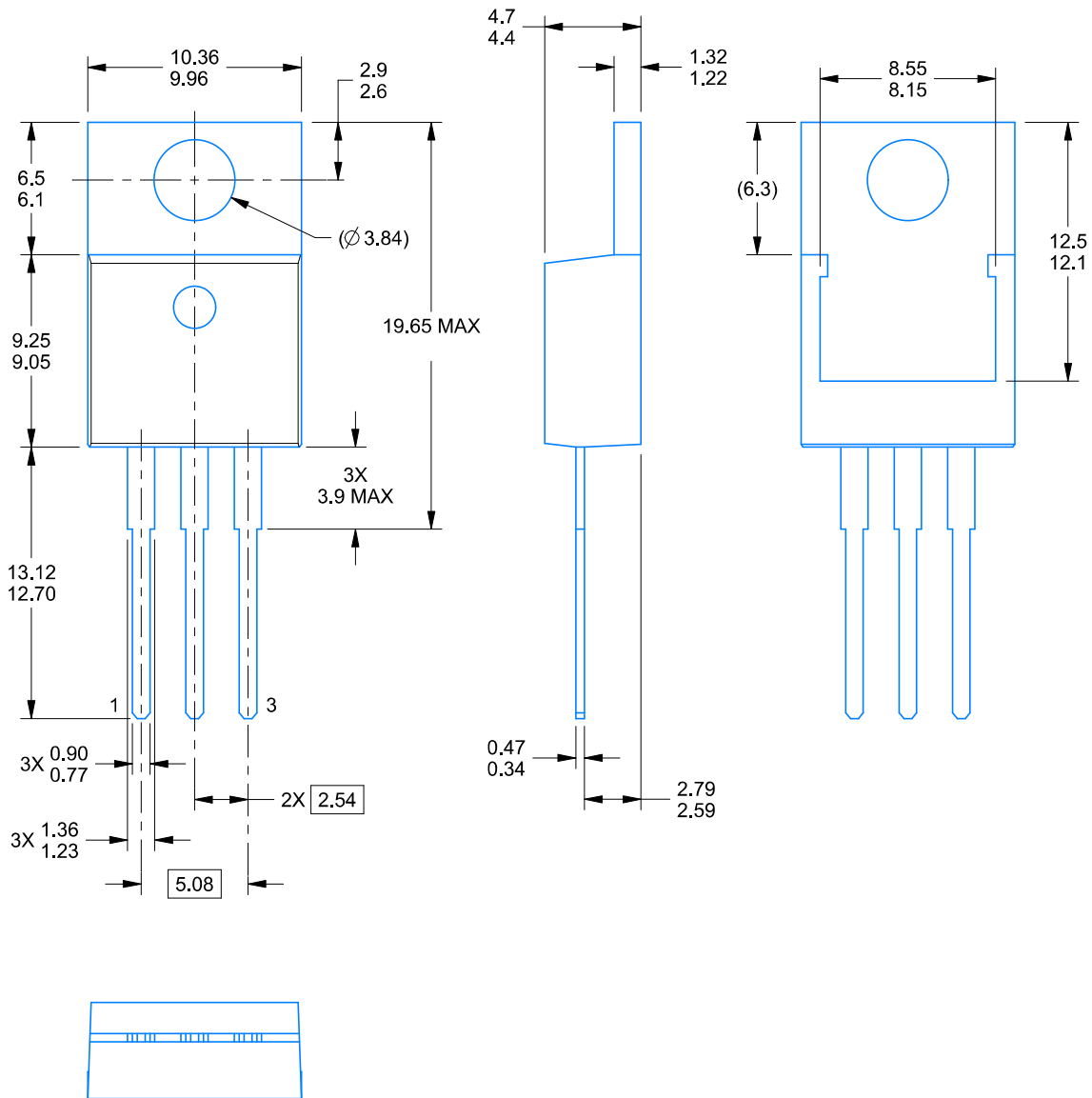
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PACKAGE OUTLINE

TO-220 - 19.65 mm max height

TO-220



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NOTES:

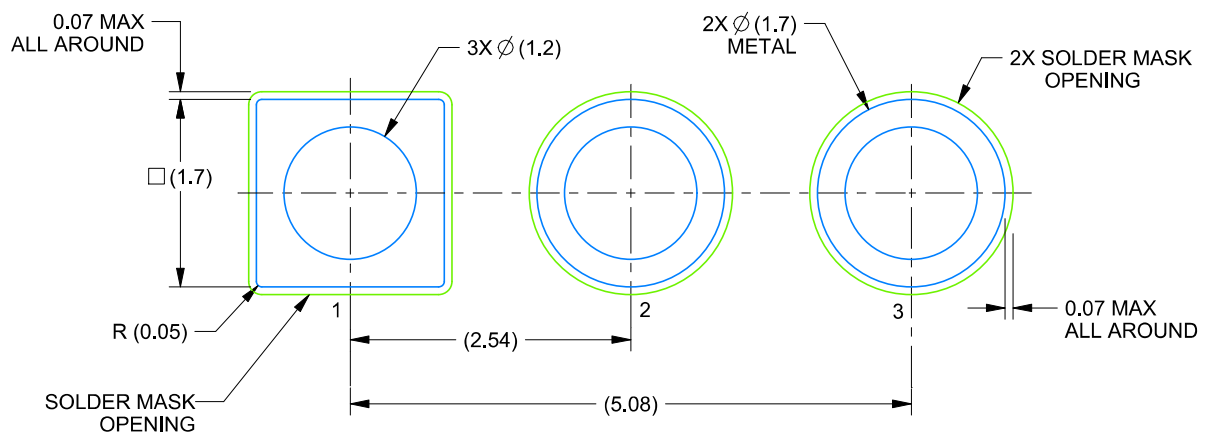
1. Dimensions are in millimeters. Any dimension in brackets or parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Reference JEDEC registration TO-220.

EXAMPLE BOARD LAYOUT

KCS0003B

TO-220 - 19.65 mm max height

TO-220



LAND PATTERN EXAMPLE
NON-SOLDER MASK DEFINED
SCALE:15X

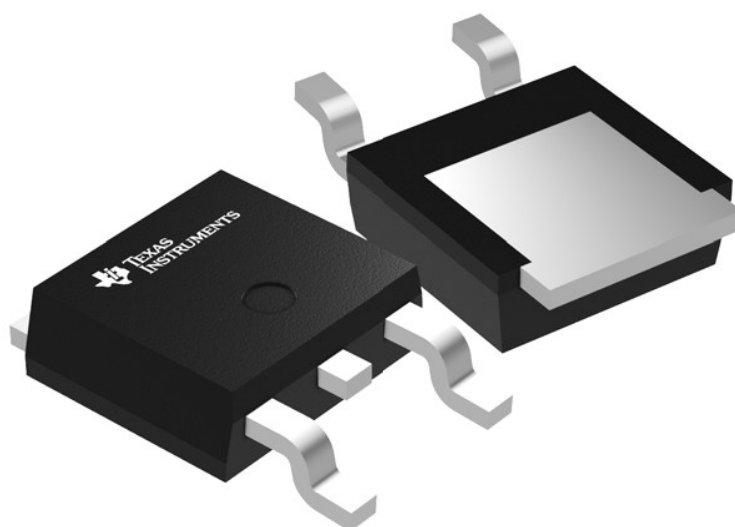
4222214/B 08/2018

KVU 3

GENERIC PACKAGE VIEW

TO-252 - 2.52 mm max height

TRANSISTOR OUTLINE



Images above are just a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

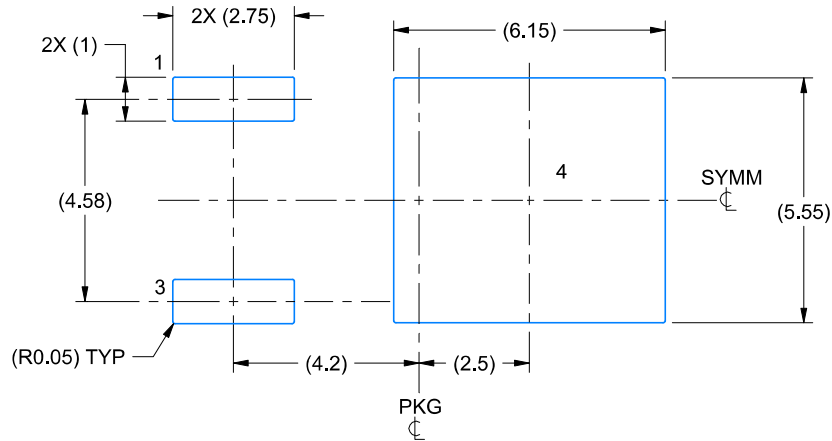
4205521-2/E

EXAMPLE BOARD LAYOUT

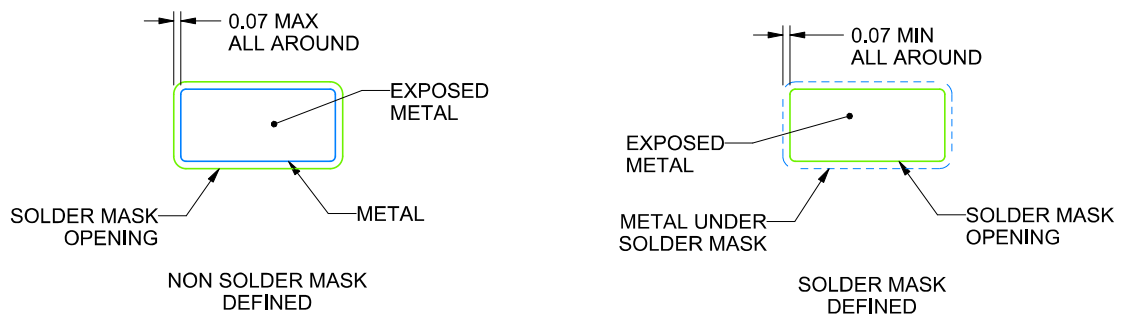
KVU0003A

TO-252 - 2.52 mm max height

TO-252



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:6X



SOLDER MASK DETAILS
NOT TO SCALE

4218915/A 02/2017

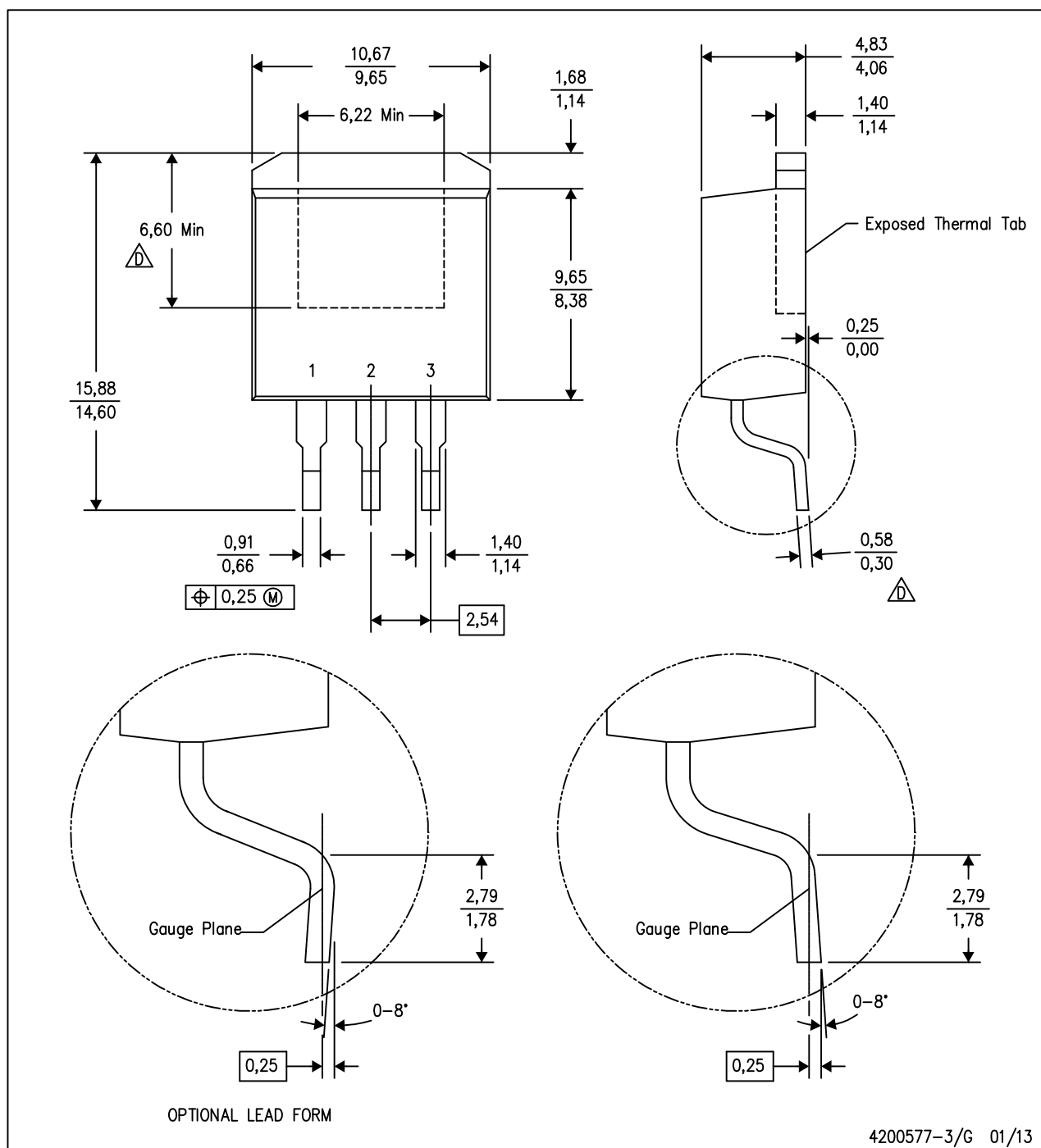
NOTES: (continued)

5. This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature numbers SLMA002(www.ti.com/lit/slm002) and SLMA004 (www.ti.com/lit/slma004).
6. Vias are optional depending on application, refer to device data sheet. It is recommended that vias under paste be filled, plugged or tented.

MECHANICAL DATA

KTT (R-PSFM-G3)

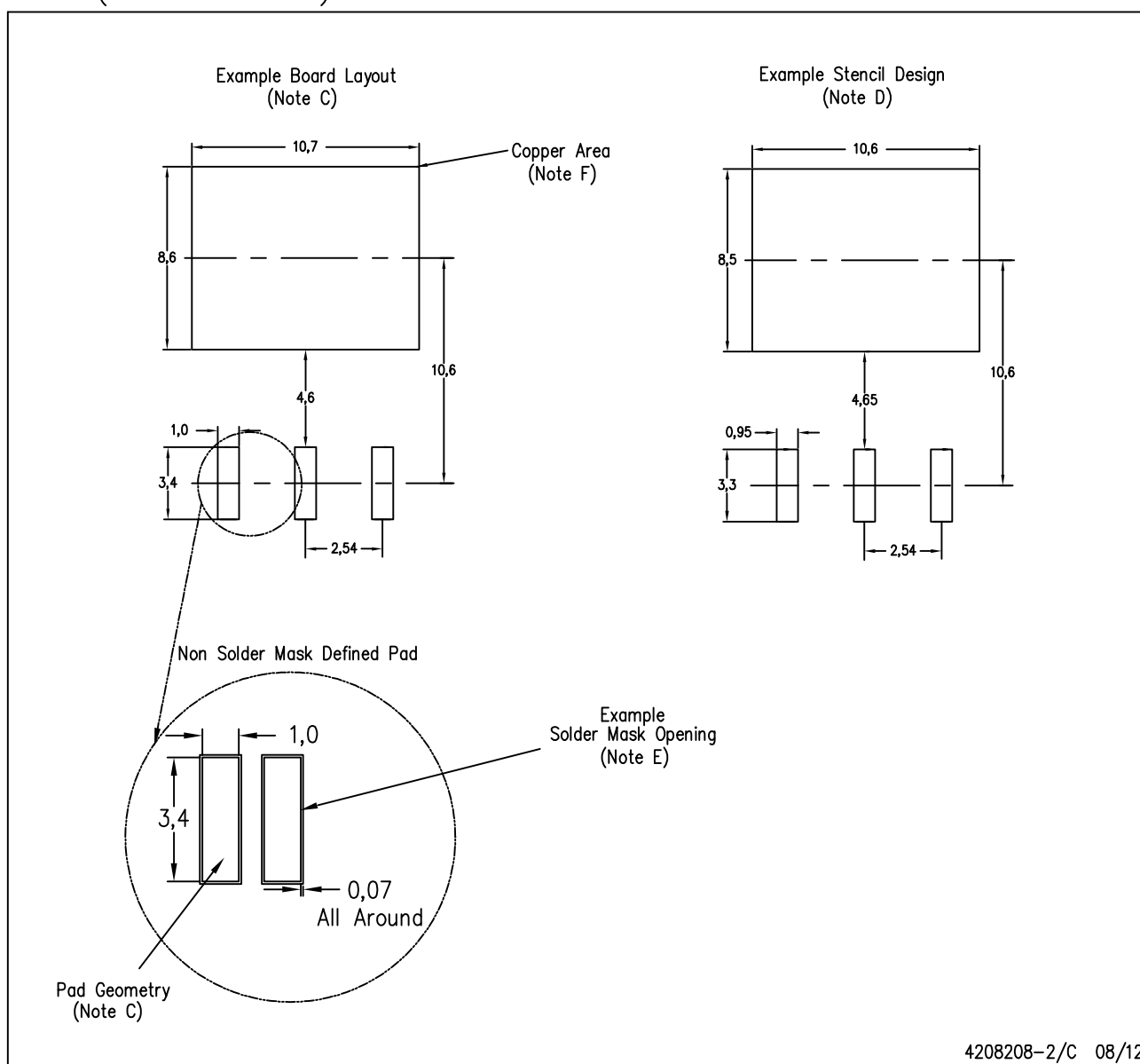
PLASTIC FLANGE-MOUNT PACKAGE



- NOTES:
- A. All linear dimensions are in millimeters.
 - B. This drawing is subject to change without notice.
 - C. Body dimensions do not include mold flash or protrusion. Mold flash or protrusion not to exceed 0.005 (0,13) per side.
 - Falls within JEDEC TO-263 variation AA, except minimum lead thickness and minimum exposed pad length.

KTT (R-PSFM-G3)

PLASTIC FLANGE-MOUNT PACKAGE



- NOTES:
- A. All linear dimensions are in millimeters.
 - B. This drawing is subject to change without notice.
 - C. Publication IPC-SM-782 is recommended for alternate designs.
 - D. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC-7525.
 - E. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.
 - F. This package is designed to be soldered to a thermal pad on the board. Refer to the Product Datasheet for specific thermal information, via requirements, and recommended thermal pad size. For thermal pad sizes larger than shown a solder mask defined pad is recommended in order to maintain the solderable pad geometry while increasing copper area.

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